

Lesson 3-1: Understand Ratios

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute 8×7 .

TASK 2 · FOUNDATION 2

Continue the pattern: 6, 12, 18, ____.

TASK 3 • FOUNDATION 3

There are 5 blue counters and 7 red counters.
How many counters in all?

TASK 4 · FOUNDATION 4

Three of eight equal slices are shaded. Write the shaded fraction.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-1: Understand Ratios

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

6×4 ? There are 3 dogs and 5 cats — how many animals in all?

TASK 6 • QUICK CHECK B

7×3 ? There are 4 red marbles and 6 blue — how many marbles in all?

TASK 7 • QUICK CHECK C

8×5 ? A team has 5 guards and 7 forwards — how many players in all?

TASK 8 • TRANSFER & EXTEND

All 28 students in a class wear either sneakers or boots. The ratio wearing sneakers to boots is 5 : 2. How many wear sneakers?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-2: Rates and Unit Rates

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $168 \div 12$.

TASK 2 • FOUNDATION 2

Five notebooks cost \$15. What is the cost in dollars for one notebook?

TASK 3 • FOUNDATION 3

A table gives 1 bag $\rightarrow$ 4 marbles, 2 bags $\rightarrow$ 8, 3 bags $\rightarrow$ 12. How many marbles for 5 bags?

TASK 4 • FOUNDATION 4

A cyclist rides 24 miles in 2 hours. State the unit rate with its unit.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-2: Rates and Unit Rates

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

$60 \div 4$? If 4 pencils cost \$2, what does 1 pencil cost?

TASK 6 • QUICK CHECK B

$72 \div 6$? If 6 notebooks cost \$9, what does 1 notebook cost?

TASK 7 • QUICK CHECK C

$150 \div 5$? If a car travels 150 miles in 5 hours, what is the speed per hour?

TASK 8 • TRANSFER & EXTEND

Brand A is 12 oz for \$3.00. Brand B is 20 oz for \$4.60. Which is the better buy?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-3: Equivalent Ratios Using Tables

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Scale the pair 3 : 5 by a factor of 4.

TASK 2 • FOUNDATION 2

Compute 9×8 .

TASK 3 • FOUNDATION 3

Fill in the numerator: $\frac{2}{3} = \frac{\quad}{15}$.

TASK 4 • FOUNDATION 4

A table lists (2, 6), (4, 12), (6, 18). What pair comes next if the same step continues?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-3: Equivalent Ratios Using Tables

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

2×5 and 3×5 ? Complete: $2 : 3 = 8 : \underline{\hspace{1cm}}$

TASK 6 • QUICK CHECK B

4×6 and 5×6 ? Complete: $4 : 5 = 24 : \underline{\hspace{1cm}}$

TASK 7 • QUICK CHECK C

3×7 and 8×7 ? Complete: $3 : 8 = 21 : \underline{\hspace{1cm}}$

TASK 8 • TRANSFER & EXTEND

A recipe uses 3 cups flour to 2 cups water. You have 8 cups of water. How much flour?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-4: Equivalent Ratios Using Graphs

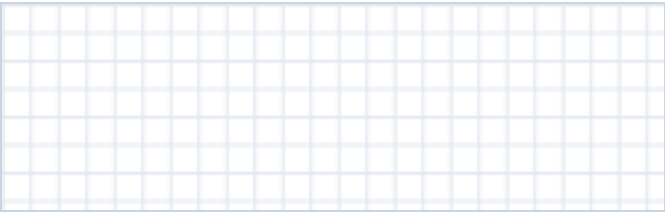
Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Describe how to plot (3, 5) from the origin.



TASK 2 · FOUNDATION 2

The x-axis ticks are 0, 2, 4, 6, 8. What x-value is at the third tick to the right of 0?

TASK 3 · FOUNDATION 3

A point is 4 units right and 7 units up from the origin. Give its ordered pair.

TASK 4 · FOUNDATION 4

The rule is $y = 3x$. Give y when $x = 4$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-4: Equivalent Ratios Using Graphs

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Start at (0, 0), move 3 units right and 2 units up — write the ordered pair. A horizontal axis has grid lines every 5 units, starting at 0. How many intervals right of 0 is the value 15?

TASK 6 • QUICK CHECK B

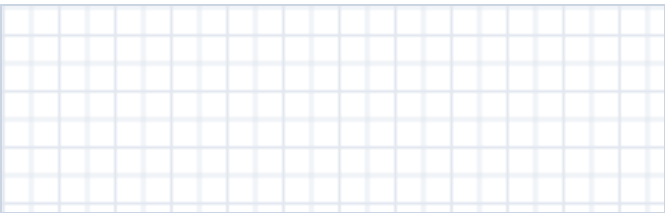
Start at (0, 0), move 5 units right and 4 units up — write the ordered pair. A horizontal axis has grid lines every 2 units, starting at 0. How many intervals right of 0 is the value 12?

TASK 7 • QUICK CHECK C

Start at (0, 0), move 2 units right and 7 units up — write the ordered pair. A horizontal axis has grid lines every 10 units, starting at 0. How many intervals right of 0 is the value 50?

TASK 8 • TRANSFER & EXTEND

A ratio graph passes through (0, 0), (2, 6), and (4, 12). What is the unit rate, and where is $x = 5$?



One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-5: Comparing Ratio Relationships

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.b

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Which is greater: 35 or 53?

TASK 2 · FOUNDATION 2

Scale 4 : 7 to an equivalent ratio with first term 20.

TASK 3 · FOUNDATION 3

What is the least common multiple of 6 and 8?

TASK 4 · FOUNDATION 4

Pack A costs \$12 for 4 pens. Pack B costs \$14 for 7 pens. Which has the lower price per pen?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-5: Comparing Ratio Relationships

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.b

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Which is greater, 12 or 9? Scale 3 : 4 so the first number is 12.

TASK 6 • QUICK CHECK B

Which is greater, 18 or 15? Scale 2 : 5 so the first number is 18.

TASK 7 • QUICK CHECK C

Which is greater, 24 or 30? Scale 4 : 7 so the first number is 24.

TASK 8 • TRANSFER & EXTEND

Runner A covers 3 miles in 24 minutes; Runner B covers 5 miles in 45 minutes. Who is faster?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-6: Converting Within One System

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $72 \div 9$.

TASK 2 • FOUNDATION 2

Use 1 foot = 12 inches. How many inches are in 3 feet?

TASK 3 • FOUNDATION 3

How many meters are in 2.4 kilometers? Use 1 km = 1000 m.

TASK 4 • FOUNDATION 4

Convert 48 inches to feet using 12 inches = 1 foot.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-6: Converting Within One System

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.d

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

12 × 4? 1 foot = ____ inches? 2.5 m = ____ cm?

TASK 6 • QUICK CHECK B

11 × 4? 1 pound = ____ ounces? 3.5 m = ____ cm?

TASK 7 • QUICK CHECK C

9 × 6? 1 gallon = ____ quarts? 0.75 km = ____ m?

TASK 8 · TRANSFER & EXTEND

A tank holds 3 gallons. How many cups is that? (1 gal = 4 qt, 1 qt = 4 cups)

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-7: Converting Between Systems

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.d

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute 6×2.5 .

TASK 2 · FOUNDATION 2

Round 7.846 to the nearest hundredth.

TASK 3 • FOUNDATION 3

Use 1 inch = 2.54 centimeters. Convert 5 inches to centimeters.

TASK 4 · FOUNDATION 4

Use 1 yard = 3 feet and 1 foot = 12 inches.
Convert 2 yards to inches.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-7: Converting Between Systems

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.d

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

4×1.6 ? 10×0.45 ? Round 6.44 to the nearest tenth.

TASK 6 • QUICK CHECK B

5×2.54 ? 8×0.45 ? Round 12.7 to the nearest tenth.

TASK 7 • QUICK CHECK C

3×1.6 ? 20×0.45 ? Round 9.08 to the nearest tenth.

TASK 8 • TRANSFER & EXTEND

A race is 5 kilometers. About how many miles is that, if $1 \text{ km} \approx 0.62 \text{ mi}$? Round to the nearest tenth.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-8: Solve Problems with Unit Rates

Connect & apply • 8 tasks • Prepares for Standard 6.RP.A.3.b

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $\$7.20 \div 6$. Enter the value only.

TASK 2 • FOUNDATION 2

Which is greater, 0.4 or 0.35? Enter the greater number.

TASK 3 • FOUNDATION 3

Compute 8×0.65 .

TASK 4 • FOUNDATION 4

A cyclist rides 36 miles in 3 hours. How many miles per hour is that?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-8: Solve Problems with Unit Rates

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.b

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

\$4.50 ÷ 6? Which is less: 0.75 or 0.8? 7×0.35 ?

TASK 6 · QUICK CHECK B

\$8.00 ÷ 10? Which is less: 0.8 or 0.75? 5×0.45 ?

TASK 7 · QUICK CHECK C

\$5.25 ÷ 3? Which is less: 1.6 or 1.75? 4×1.25 ?

TASK 8 · TRANSFER & EXTEND

A printer prints 56 pages in 8 minutes. How many pages does it print in 20 minutes at the same rate?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-9: Equivalent Ratios

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute 7×6 .

TASK 2 · FOUNDATION 2

Find the greatest common factor of 18 and 24.

TASK 3 • FOUNDATION 3

Find the missing factor: $9 \times ? = 63$.

TASK 4 · FOUNDATION 4

A class has 12 girls and 15 boys. Write the ratio of boys to girls and explain which term comes first.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 3-9: Equivalent Ratios

Connect & apply · 8 tasks · Prepares for Standard 6.RP.A.3.a

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

15 ÷ 5? GCF of 15 and 25? 4 × ? = 28?

TASK 6 • QUICK CHECK B

18 ÷ 6? GCF of 12 and 18? 3 × ? = 24?

TASK 7 • QUICK CHECK C

36 ÷ 9? GCF of 16 and 40? 6 × ? = 42?

TASK 8 · TRANSFER & EXTEND

A bread recipe uses 5 cups of flour to 2 cups of milk. A bakery batch uses 14 cups of milk. How much flour keeps the same ratio?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

